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# When does social desirability become a problem? Detection and reduction of social desirability bias in information systems research

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A B S T R A C T

Social desirability (SD) bias occurs in self-report surveys when subjects give socially desirable responses by over- or underreporting their behavior. Despite knowledge of SD as a potential threat to the validity of information systems (IS) research, little has been done to systematically assess its extent. Furthermore, we are uncertain of how to recover reliable estimates of the relationships between research variables contaminated by SD bias. We sought in this study to assess the extent of SD bias in causal inferences when independent and/or dependent variables are contaminated. We also evaluated whether an SD scale in conjunction with partial correlation could effectively and efficiently correct SD bias when it is found. To achieve these purposes, we designed a survey study and collected data from Amazon's Mechanical Turk in the context of mobile loafing, which refers to employees' personal use of the mobile Internet during business hours. Using various detection methods, we found that SD bias existed in the context of mobile loafing. From the results of the variance reduction rate and a covariate technique, we found that SD bias becomes problematic when both the independent and dependent variables are susceptible to SD bias. Overall, our study contributes significantly to the IS literature by revealing the extent of SD bias and the magnitude of the possible correction for it in IS research.

## 1. Introduction

Surveys play a critical role in Information Systems (IS) research. A key assumption of self-report surveys is that respondents bring to mind appropriate information accurately and respond honestly [1]. Social desirability (SD) bias, one of the most frequently raised concerns with self-report surveys, is a tendency for subjects to give socially desirable responses by overreporting behaviors (e.g., knowledge contribution) that make them look good and underreporting behaviors (e.g., cyberbullying) that make them look bad [2–4]. This bias is problematic because it distorts the means and variances of research variables and threatens the validity of survey-based causal inferences in IS research [5,6]. For these reasons, conclusions drawn from analyses using potentially biased self-report data can misguide managerial and policy decisions [6,7]. Thus, it is crucial for researchers and practitioners to assess the extent of SD bias and, if necessary, control for it in drawing inferences from their survey data.

Recent IS studies have attempted to assess the extent of SD bias when researchers examine sensitive topics such as addiction, software piracy, unethical programming, and cyberbullying [5,6,8–18]. Notable methods to assess and/or reduce SD bias in prior IS research are (1) indirect questioning, (2) a randomized response technique (RRT), (3) implicit constructs and measures, and (4) the SD scale and covariate techniques. First, indirect questioning is a projective technique in which subjects are asked to report the thoughts or actions of others similar to themselves [19]. Second, the RRT seeks to prevent SD bias during data collection by protecting privacy through a randomization device with a known probability distribution (e.g., coins, dice) [6,20]. Third, implicit constructs that are only indirectly measured can reduce SD bias because implicit constructs do not depend on explicit and direct measures. Fourth, researchers can include an SD scale in a survey questionnaire and then assess SD bias by examining the correlations between the SD scale and research variables [9,21]. If significant correlations are found, a covariate technique using partial correlation can be used to reduce SD bias [22–24].

Previous IS researchers have relied on one or other of the alternative

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approaches to assess the extent of SD bias, but their conclusions have been contradictory. For example, some research, based on the relatively low correlations between an SD scale and sensitive variables, indicated SD bias was not a problem [8,13–15,17]. In contrast, Kwan et al. [6] found the structural paths of respondents with direct questioning were significantly higher than those of respondents with the RRT and thus concluded that SD bias is a real threat in IS research. Similarly, in social psychology, some researchers considered SD bias significant, but others believed it to be trivial [25]. These contradictory conclusions generally hinged on the context and contingencies of admitting to undesirable behavior. For example, anger measures from parolees are significantly susceptible to SD bias [26], whereas admitting anger among college students is not [27].

To resolve these contradictions about SD bias, Paunonen and LeBel [25] used a Monte Carlo simulation to evaluate its effects in an independent variable on an unbiased dependent variable. They concluded that SD bias is not seriously problematic, at least not in a scenario in which only one variable is biased. Nevertheless, despite their findings, the extent of SD bias remains uncertain when both independent and dependent variables are susceptible to it. Although various methods to detect this bias have been proposed, prior IS research has relied heavily on correlation analysis that used a short form of the Marlowe–Crowne SD scale. Furthermore, we remain unsure how to effectively control for SD bias, especially when the RRT is not viable. It is especially important to understand how effective the technique of using SD scales could be in recovering reliable estimates of the relationships between contaminated research variables.

Our main objective in this study was twofold: First, it seeks to assess the extent of SD bias in the estimation of causal relationships when it contaminates independent and/or dependent variables. Second, it evaluates the effectiveness of SD scales with partial correlations in controlling SD bias and acquiring reliable estimates of causal relationships. To achieve these goals, we designed a survey study and collected data from Amazon's Mechanical Turk (MTurk) in the context of mobile loafing, or "nonwork-related mobile computing," which refers to employees' personal use of the mobile Internet during business hours [28–31]. Our survey is intended to examine how the estimation of causal relationships varies depending on the different conditions of SD bias in independent and dependent variables. This study used indirect questioning, correlation analysis, and a factor mixture model (FMM) to assess the extent of SD bias. It was also designed to help understand how useful a covariate technique is in recovering true causal relationships free of SD bias. To correct for SD bias, we used the methods of a variance reduction rate (VRR) and a covariate technique.

This study contributes to IS research in several ways. First, by venturing beyond the one-biased-variable condition examined in Paunonen and LeBel [25], we contribute to the literature by examining a two-biased-variable condition. Our contribution lies in our finding that, although the level of distortion in causal inferences is minor in a one-biased-variable condition—which is consistent with the prior literature—a two-biased-variable condition may cause a problematic degree of change in causal inferences. Second, our results can resolve the different conclusions of the effects of SD bias in previous IS research. In particular, a one-biased-variable condition supports the argument that SD bias is not problematic (e.g., [13,15]), but a two-biased-variable condition supports the conclusion that SD bias may threaten the validity of self-report research (e.g., [6]). Third, unlike prior IS research that relied on a single method to assess SD bias, our study used—and evaluated—multiple methods. Among these various methods, we found that SD scales are more useful than other techniques because they can assess the degree of SD bias and control it by using partial correlations. Finally, we found that the impression management scale is efficient and effective in measuring and controlling for SD bias in the contexts of negative use of information technology (IT) such as addictive IT use, cyberbullying, and digital piracy.

This article is organized as follows. In the next section, we review SD bias and techniques for reducing it. Then, we present an empirical model for assessing and controlling SD bias and a discussion of our methods. Next, we examine four methods to detect SD bias and a covariate technique to reduce it. Finally, we conclude with theoretical, methodological, and practical contributions, along with limitations of our study and directions for future research.

## 2. Literature review

### 2.1. Social desirability bias

Individuals prone to SD bias respond to questions in ways to make themselves viewed favorably by overreporting socially desirable behaviors or underreporting socially undesirable behaviors. SD has been arguably viewed as either a substance (personality trait) or style (characteristic of survey items) [7,32]. In other words, SD can be seen as an individual difference variable or a property of survey items [33]. Most of the existing SD scales are aimed at capturing substance rather than style [33,34]. A few studies have also concluded that when major personality variables are controlled for in various contexts, the impact of SD on the dependent variables in those contexts can be removed [32]. Because much prior research has concluded that SD is a type of personality, our research focuses on the personality aspect of SD, that is, subjects' tendency to give positive self-descriptions [35].

Variables that are socially desirable can covary, not because they are associated, but because both of them are contaminated by SD bias [33]. Accordingly, SD bias poses a severe problem because it threatens the validity of causal inferences in research with self-report surveys [6]. However, the issues caused by SD bias can be diagnosed and resolved if a researcher can measure the extent to which each subject provides his or her answers in a socially desirable way [4,33]. For example, high correlations between an SD scale and study variables suggest the scale has a validity issue and that a partial correlation using an SD scale can remove SD bias [23,24].

Several SD scales have been developed to measure individual differences in SD (see [4] for a review). They are the Edwards SD scale [36], the Wiggins Sd scale [37], the Marlowe–Crowne (MC) scale [38], and the balanced inventory of desirable responding (BIDR) [39,40]. The early SD scales (e.g., Edward's, Wiggins's, and the MC) were developed based on a unidimensional conceptualization of socially desirable responding. However, low correlations between these early scales led to the formulation of two-factor models [7,39,41]. Based on the multidimensionality of SD, Paulhus [39] developed the BIDR. It consists of two subscales: (1) the self-deception management (SDE), which assesses subjects' tendencies to attribute positive characteristics to themselves, and (2) the IM scale, which measures a subject's tendency to give a socially positive self-image to others.

Although the MC scale and the BIDR are the two most widely used SD scales, prior IS research has predominantly used a short form of the MC scale [42] to evaluate the extent of SD bias (e.g., [8,9,13–15,17]). However, because the MC scale and its short forms have several limitations (e.g., including insensitive and outdated items, unidimensionality) [3,43], the BIDR is now considered the best way to assess the extent of SD bias [7,44]. Thus, we used the BIDR to evaluate and control SD bias in the context of mobile loafing.

### 2.2. Reduction methods of SD bias

According to Paulhus [24], methods for controlling SD bias can be divided into three categories: "(1) rational techniques—controls built into the survey instrument to prevent SD bias, (2) principal factor deletion—the deletion of SD contamination during the process of deriving factors from an item-correlation matrix, and (3) covariate techniques—the removal of SD contamination from scale scores using partial correlation or regression approaches" (pp. 383–384). To represent the use of rational techniques, we relied on indirect questioning and the RRT because they were used in prior IS research.

Indirect questioning is a projective technique in which subjects are asked to respond from the perspective of others similar to themselves [19,45]. Accordingly, indirect questioning (e.g., a typical Instagram user is addicted to Instagram) can mitigate SD bias and elicit responses less susceptible to SD bias than can be done with direct questioning (e.g., I am addicted to Instagram). Prior IS research has used indirect questioning to reduce SD bias [46,47] and to assess the presence of SD bias [5]. Despite its benefits, indirect questioning has two major limitations. First, when a respondent projects himself or herself onto what is presumably typical of others, the answer still retains a degree of SD bias. Second, the answer is not directed toward a specific person under study but instead toward others in general. As a result, the data collected through indirect questioning could introduce other biases [20].

The central idea in RRT is "to assure complete confidentiality of a subject's response by contaminating it with random noise with known statistical properties" ([6], p. 942). For its random noise, the RRT uses randomization devices (e.g., cards, dice) to generate probability distribution [20,48]. In IS research, Kwan et al. [6] extended the RRT to multivariate analysis. They examined software piracy behavior and found that subjects responding to RRT were likelier to disclose sensitive information about their software piracy than they were with direct questioning. Although the RRT can effectively prevent SD bias, it is considered less efficient than direct questioning because it suffers "from larger standard errors, leading to reduced power which makes it necessary to use larger samples" ([49], p. 254). Moreover, because the randomization procedure associated with coin flipping or dice rolling is difficult to administer, RRT is less practical than other approaches. After Kwan et al. [6] introduced RRT to IS research, no IS research has used the RRT to prevent SD bias.

An implicit construct is a stable concept of an object "that is formed a-priori, is stored in special fast-access memory, and is activated with little or no conscious effort in response to internal or external stimuli" related to the object ([50], pp. 657–658). Because survey-based measures are explicit in their nature, SD bias can be mitigated using implicit constructs by indirectly measuring them [50]. Prior IS research used the implicit association test (IAT) as a technique for implicit construct measurement [50,51]. Despite its potential to reduce SD bias, implicit constructs require expertise to design, administer, and analyze IAT (see [51] for detailed information about IAT).

The principal-factor deletion technique was introduced by Paulhus [24]. The technique is used during factor analysis when derived factor scales are expected to be contaminated by SD bias. If SD is found to be represented by one principal factor, the SD-contaminated factor is removed. Then, before the remaining factors are rotated to the desired factor structure, communalities are adjusted. Subscales stemming from resulting factors are expected to be immune to SD bias. This technique was applied to an inventory of the locus of control items [52]. However, this is an old method that dates back to when researchers made a program to run exploratory factor analysis. Thus, it is difficult to follow the procedure because modern researchers use off-the-shelf products such as SPSS.

Finally, covariate techniques using SD scales can be used when a researcher finds significant correlations between research variables and an SD scale. This technique makes it possible to control for SD bias through partial correlations [4]. Because of their effectiveness and convenience, SD scales have been used more than other alternatives. In particular, prior IS researchers have mostly used a short form of the MC scale and concluded that SD bias was not problematic because of low correlations (e.g., [9,13–15,17]). However, the results from the MC scale are inconclusive because it is ineffective in assessing SD bias [7]. Thus, we used the BIDR and then a covariate technique to derive the accurate relationships if significant correlations were found.

## 3. Empirical study

### 3.1. Mobile loafing as an empirical setting

To evaluate the extent of SD bias under different conditions, we collected survey data in the context of mobile loafing [28–31]. The widespread diffusion of personal mobile devices (e.g., smartphones, tablets) has multiplied the situations and locations that permit individuals to access the Internet. As of January 2021, the number of global mobile Internet users was 4.32 billion, which is more than the number of desktop Internet users [53]. Employees' use of the mobile Internet is a double-edged sword. Some studies have argued that its use for work can increase employee engagement, productivity, and customer service [54,55].

However, others have branded mobile loafing as a deviant workplace behavior (i.e., socially undesirable) because it reduces employees' engagement and productivity and makes companies more vulnerable to security threats [29,56–58]. Although it is a minor type of cyberloafing [59], mobile loafing is considered counterproductive workplace behavior. As a result, people are likely to distort their responses to sensitive questions in a socially desirable manner. As a result, mobile loafing is a proper context for studying the extent of SD bias. The following section presents a conceptual model of mobile loafing before we describe the measurement items and data collection.

### 3.2. Empirical model and constructs

Fig. 1 shows the model examined in this study. Our empirical model included four constructs: mobile Internet addiction, perceived usefulness, neutralization, and mobile-loafing intention. We expected that mobile Internet addiction, neutralization, and mobile-loafing intention were susceptible to SD bias, but perceived usefulness was free of SD bias. In other words, mobile Internet addiction, neutralization, and mobile-loafing intention would have negative correlations with SD scales, but perceived usefulness would show an insignificant correlation with SD scales. Our rationales are provided below:

Mobile Internet addiction refers to "a psychological state of maladaptive dependency on the use" of the mobile Internet ([15], p. 1044). The addiction covers a range of contexts from substance addictions (e.g., alcohol, tobacco, drugs) to behavioral addictions (e.g., gambling, sex, food, shopping, work, tanning, exercise) [60,61]. Recently, another form of behavioral addiction, the use of IT (e.g., video games, smartphones, social networking sites), has received significant attention [61,62]. Thus, technology addiction such as mobile Internet addiction can disrupt cognitive and emotional functioning, compromise social relations, and inhibit academic and workplace performance [14,61]. Because addiction is a sensitive issue, prior research on technology addiction assessed the extent of SD bias in the context of auction addiction [15], social networking website addiction [14], and children's Internet addiction [17]. Accordingly, respondents would underreport their perceptions of mobile Internet addiction.

Neutralization is defined as the degree to which employees attempt to excuse, rationalize, or justify their mobile-loafing behavior [30]. Neutralization encourages deviant behaviors and motives by reducing a person's sense of guilt and concern for others' disapproval of their negative actions [29,63]. According to the neutralization theory, regardless of whether people follow the rules or not, they believe in society's norms and values [64,65]. Neutralizing mobile loafing may not be powerful enough to justify and rationalize employees, making them feel guilty and shameful [65]. It merely means that employees who neutralize mobile loafing acknowledge that they conform to social values and may look bad. Thus, employees with SD bias are likely to score low on neutralization.

Mobile-loafing intention refers to employees' likelihood to use the mobile Internet for nonwork-related reasons at the workplace [29]. As mentioned earlier, mobile loafing is considered a problematic workplace

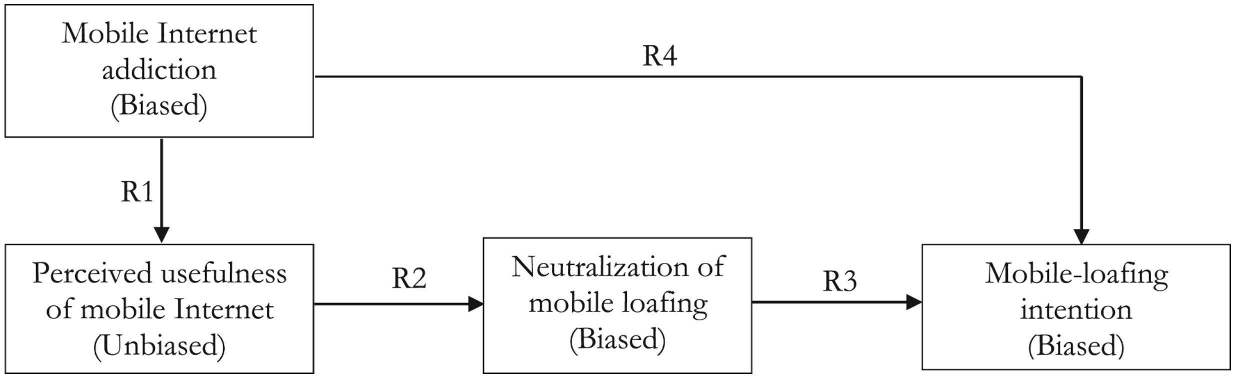


**Fig. 1.** Empirical model.

behavior because it dampens employees' engagement and productivity [28,56]. Consequently, employees who are willing to commit mobile loafing will underreport their mobile-loafing intention because mobile loafing will make them look bad.

Finally, we expect perceived usefulness of the mobile Internet to be free of SD bias. We defined perceived usefulness as the degree to which an employee believes that using the mobile Internet would increase his or her performance at work [66]. Perceived usefulness is not prone to SD bias because people typically feel free to express their opinions honestly and candidly when evaluating a neutral object (i.e., mobile Internet). Accordingly, individuals have little need to hide their genuine feelings about the utility of the mobile Internet. Prior research found an insignificant relationship between perceived usefulness and SD scales, suggesting that perceived usefulness is not susceptible to SD bias [15].

### 3.3. *Causal relationships*

Turel et al. [15] provided several psychological mechanisms that can explain the effect of addiction on positive perceptions of a system. For instance, a key tenet of a sequential updating mechanism is that individuals' perceptions of a postusage stage are influenced by perceptions formed at the prior usage stage [67–69]. Because addicted people have higher positive evaluations of a system [70], their inflated evaluations will impose a higher anchoring point for perceptions of usefulness [15]. Also, Turel et al. [15] found that auction addiction positively influences perceived usefulness, ease of use, and enjoyment. Thus, we test the following relationship:[1]

> Relationship 1: Mobile Internet addiction is positively associated with perceived usefulness of the mobile Internet.

We expect perceived usefulness of the mobile Internet at work to have a positive effect on neutralization of mobile loafing. It is noteworthy that perceived usefulness is defined in a work-related context (e.g., improving performance), whereas mobile loafing is defined in a nonwork-related context (e.g., gaming). The relationship between these two different contexts can be explained by confirmation bias, which refers to people's tendencies to seek and use information that confirms their existing beliefs [71,72]. After positive interactions with the mobile Internet, employees may find confirmatory evidence and logical justification for their personal use of the mobile Internet during business hours. Therefore, we examine

> Relationship 2: Perceived usefulness of the mobile Internet is positively associated with neutralization of mobile loafing.

Criminology theories such as social learning theory [73] argue that people's intentions to commit deviant behaviors are affected by their tendencies to neutralize the behavior. Similarly, prior IS research has found that people use neutralization to justify their intention to commit problematic workplace behaviors in the context of information security violations [63,74], software piracy [75], and cyberloafing [29–31]. The findings of prior research will apply to the context of mobile loafing. Thus, we examine

> Relationship 3: Neutralization of mobile loafing is positively associated with mobile-loafing intention.

Finally, we tested the relationship between mobile Internet addiction and mobile-loafing intention. Prior research suggested various addiction symptoms [15,76–78]. For example, addicted people's thoughts and behaviors are dominated by Facebook, and thus, they are unable to reduce Facebook use voluntarily [78,79]. Also, negative emotions such as stress or anxiety arise if a person cannot use Facebook [79,80]. Subsequently, the use of Facebook conflicts with other duties in the workplace [81]. Based on the above-mentioned symptoms of addiction, we expect that employees who are addicted to the mobile Internet are likely to commit mobile loafing.

> Relationship 4: Mobile Internet addiction is positively associated with mobile-loafing intention.

## 4. Methods

### 4.1. *Measures*

To ensure construct validity, all measures were adapted from previously validated scales. The scales used in our study are included in the Appendix. The survey included direct questions, indirect questions, SD scales, and demographic questions. Mobile Internet addiction was adapted from Kuem et al. [82]. The scales for perceived usefulness were adapted from Davis [66]. We measured neutralization and mobile-loafing intention by using scales adapted from Khansa et al. [29].

All the scales of the study constructs were measured for direct and indirect questioning by using a 7-point Likert scale anchored by *strongly disagree* and *strongly agree*. For the SD scale, we used the BIDR-16 [3] because it is shorter than the original version [4,83] and other short forms [7] while retaining reliability, validity, and a two-factor structure [3]. The BIDR-16 contains 8 items of the SDE scale and 8 items of the IM scale. The BIDR-16 was measured with a 7-point scale anchored with *not true* and *very true*. After inversing the socially undesirable questions, one point was added for each extreme response (6 or 7) [4,83]. Ranges of entire scores on the SDE scale and the IM scale were 0–8, and 0–8, respectively. Finally, the survey included demographic information such as gender, age, educational level, and frequency of mobile Internet use.

[1] We use the term "relationship" rather than "hypothesis" because our methods focus on correlation.

### 4.2. Data collection

After developing an initial version of the survey, several faculty members and doctoral students pretested it for clarity and content validity. They recommended that we include a definition of mobile loafing. Accordingly, we added at the beginning of the survey a definition of mobile loafing (mobile-loafing, or "non-work-related mobile computing," refers to employees' using mobile Internet during business hours for personal reasons (e.g., gaming, browsing websites, sending messages, etc.)). After the pretest, we conducted a field study to collect the data. We considered the population of interest to be currently employed and current mobile Internet users in the United States. For the field study, we recruited subjects from MTurk, an online crowdsourcing platform where participants complete certain online-based tasks to earn money. We selected MTurk because its population is diverse and reliable, increasing the external validity of data [84].

We recruited 214 MTurk workers currently employed and users of the mobile Internet in the United States. To test for response bias, we examined whether early and late respondents differed statistically. We found no significant difference in gender ($\chi^2 = 2.37$, $p = .12$) or age ($F = 2.01$, $p = .16$), suggesting response bias was not a serious concern. We discarded nine responses that failed to correctly answer the bogus questions (e.g., all my friends are aliens). This adjustment yielded 205 usable observations. In the final data set, the respondents' average age was 36.9, and the percentage of females was 51.7%; 67.3% of the respondents spent more than seven hours a week on the mobile Internet.

### 4.3. Measurement model

To assess the measurement quality of constructs, we conducted a confirmatory factor analysis (CFA) using AMOS 22.0. The overall fit indices of the measurement model suggested a good fit of the model because most of the indices were at or above the recommended values.[2] Fit indices of the measurement model ($\chi^2$ (307) = 447.39) are as follows: $\chi^2$/df = 1.46, the comparative fit index (CFI) = 0.97, the goodness of fit index (GFI) = 0.86, the adjusted goodness of fit (AGFI) = 0.82, the standardized root mean square residual (SRMR) = 0.038, and the root mean square error of approximation (RMSEA) = 0.047.

The measurement model was further assessed by examining the validity and reliability of the scales. First, convergent validity was established if the factor loading of an item exceeded 0.707 [85]. Second, the reliability of each construct was assessed with Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE) [85]. The recommended cut-off values for Cronbach's alpha, CR, and AVE are 0.70, 0.70, and 0.50, respectively [85,86]. As shown in Table 1, all of these values were satisfactory. Discriminant validity was assessed by comparing the square root of AVE for each construct with the correlations the construct had with other constructs [87]. The square root of the AVE for each construct exceeds its correlations with other constructs, suggesting discriminant validity. Table 1 shows correlations, means, standard deviations, reliability, and validity of study constructs.

The reliability of the SD scales was examined using Cronbach's alpha. Our results from Cronbach's alpha (SDE: 0.83, IM: 0.76) exceeded 0.70, suggesting acceptable internal consistency. We further assessed the validity of the SD scales by examining the factor structure of the BIDR. As prior research suggested, the BIDR was developed for a two-factor structure [3,7,39]. Two CFAs were performed to test a single-factor model and a two-factor model. The single-factor model exhibited a poor fit with $\chi^2 = 291.6$ (df = 104), $\chi^2$/df = 2.80, CFI = 0.76, GFI = 0.82, AGFI = 0.76, SRMR = 0.088, and RMSEA = 0.094; the two-factor model showed a better fit with $\chi^2 = 173.8$ (df = 103), $\chi^2$/df = 1.69, CFI = 0.91, GFI = 0.82, AGFI = 0.87, SRMR = 0.060, and RMSEA = 0.058. We further conducted an $\chi^2$ difference test [88]. If the difference is significant, we can conclude that the two-factor model is superior to the single-factor model. The $\chi^2$ difference test was significant ($\Delta\chi^2(1) = 117.8$, $p < .001$), suggesting that the BIDR has two distinct factors.

### 4.4. Structural model

A structural model using uncorrected correlations was developed to examine causal relationships among the constructs[3]. All the values are within an acceptable range for good model fit: $\chi^2$/df = 1.48, CFI = 0.99, GFI = 0.93, AGFI = 0.90, SRMR = 0.061, and RMSEA = 0.048. The results of the uncorrected structural model are presented in Fig. 2. Uncorrected estimates of the mobile-loafing model showed that mobile Internet addiction had significant effects on perceived usefulness ($\gamma = 0.25$, $p < .01$) and mobile-loafing intention ($\gamma = 0.14$, $p < .05$). Also, perceived usefulness positively influenced neutralization of mobile loafing ($\beta = 0.49$, $p < .001$), which, in turn, affects mobile-loafing intention ($\beta = 0.63$, $p < .001$). The results support the causal relationships of the mobile-loafing model. In the following sections, this study assesses the extent of SD bias and uses partial correlations to test the corrected structural models.

## 5. Detection of social desirability bias

To assess SD bias, we used three methods: (1) comparisons between indirect and direct questioning [19], (2) correlations between study constructs and SD scales [3,4], and (3) an FMM [89].

### 5.1. Comparison between indirect and direct questioning

Indirect questioning is a technique to lessen SD bias on self-report measures by allowing respondents to project their own beliefs/attitudes onto others they consider similar to themselves [19]. Compared with direct questioning, indirect questioning leads to higher (vs. lower) response scores for socially negative/undesirable (vs. positive/desirable) behaviors. A multivariate analysis of variance (MANOVA) test was conducted to examine the effect of questioning type (i.e., indirect vs. direct). As shown in Table 2, the results indicate a significant difference between indirect and direct questioning. We found that Wilks' lambda value of questioning type (Wilks' $\lambda = 0.76$, $p < .001$) was significant, suggesting that there is SD bias overall.

Because the multivariate significance could be due to a combination of dependent variables rather than an individual dependent variable, we conducted an analysis of variance (ANOVA) test. The test (see Table 2) showed that there are significant differences between indirect and direct questioning in mobile Internet addiction ($M_{\text{indirect}} = 4.05$, $M_{\text{direct}} = 2.55$; $F = 113.02$, $p < .001$), neutralization ($M_{\text{indirect}} = 5.18$, $M_{\text{direct}} = 4.64$; $F = 11.18$, $p < .01$), and mobile-loafing intention ($M_{\text{indirect}} = 5.27$, $M_{\text{direct}} = 4.94$; $F = 6.18$, $p < .05$). These differences suggest these constructs are susceptible to SD bias. However, we found no significant difference in perceived usefulness ($M_{\text{indirect}} = 4.39$, $M_{\text{direct}} = 4.42$; $F = 0.05$, $p > .05$), implying that it is free of SD bias. Consistent with our expectations, our results suggested that SD bias existed in the context of mobile loafing.

### 5.2. Correlation analysis

A significant correlation between an SD scale and the study's constructs suggests a certain level of SD bias [3,4,19]. Table 3 shows the

[2] Recommended cutoff values for these fit indices are as follows: Below 3.0 for $\chi^2$/df; above .90 for CFI; above .90 for GFI; above .80 for AGFI; .08 for SRMR; below or .08 for RMSEA [97–99].

[3] The structural model is based on direct questioning. We did not use indirect questioning for the structural model because indirect questioning is not directed toward a participant but instead toward others in general. Thus, indirect questioning could produce other biases [7] in examining causal relationships between variables.

**Table 1**
Correlation matrix, reliability, and validity.

| | BIDR scale | | Indirect questioning | | | | Direct questioning | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | SDE | IM | ADDi | PUi | NEUi | INTi | ADDd | PUd | NEUd | INTd |
| SDE | – | | | | | | | | | |
| IM | .446$^{***}$ | – | | | | | | | | |
| ADDi | −0.108 | −0.180$^{*}$ | **.804** | | | | | | | |
| PUi | −0.051 | −0.033 | .150$^{\dagger}$ | **.905** | | | | | | |
| NEUi | −0.037 | .073 | .096 | .268$^{***}$ | **.888** | | | | | |
| INTi | .059 | −0.119 | .298$^{***}$ | .205$^{**}$ | .553$^{***}$ | **.915** | | | | |
| ADDd | −0.333$^{***}$ | −0.325$^{***}$ | .526$^{***}$ | .177$^{*}$ | −0.036 | −0.058 | **.801** | | | |
| PUd | .020 | −0.007 | .162$^{*}$ | .602$^{***}$ | .181$^{*}$ | .207$^{**}$ | .243$^{**}$ | **.901** | | |
| NEUd | −0.083 | −0.127$^{\dagger}$ | .020 | .283$^{***}$ | .500$^{***}$ | .461$^{***}$ | .139$^{\dagger}$ | .482$^{***}$ | **.905** | |
| INTd | −0.075 | −0.211$^{**}$ | .150$^{\dagger}$ | .271$^{***}$ | .342$^{***}$ | .615$^{***}$ | .219$^{**}$ | .496$^{***}$ | .643$^{***}$ | **.958** |
| Mean | 2.830 | 2.820 | 4.049 | 4.385 | 5.184 | 5.272 | 2.549 | 4.419 | 4.644 | 4.935 |
| S.D. | 2.531 | 2.293 | 1.355 | 1.543 | 1.472 | 1.404 | 1.498 | 1.474 | 1.782 | 2.029 |
| Alpha | .826 | .762 | .843 | .878 | .930 | .917 | .939 | .875 | .926 | .931 |
| CR | – | – | .879 | .931 | .918 | .939 | .877 | .928 | .931 | .971 |
| AVE | – | – | .646 | .819 | .789 | .837 | .642 | .812 | .819 | .917 |
| Factor loadingranges | – | – | .758-.846 | .866-.942 | .844-.921 | .894-.934 | .761-.851 | .840-.933 | .857-.929 | .914-.982 |

Notes:
SDE: Self-deception enhancement; IM: Impression management; ADDi: Mobile Internet addiction – indirect questioning; PUi: Perceived usefulness of mobile Internet – indirect questioning; NEUi: Neutralization of mobile loafing – indirect questioning; INTi: Mobile-loafing intention – indirect questioning; ADDd: Mobile Internet addiction – direct questioning; PUd: Perceived usefulness of mobile Internet – direct questioning; NEUd: Neutralization of mobile loafing – direct questioning; INTd: Mobile-loafing intention – direct questioning
A two-tailed test was used.
*Bold*: Square root of AVE.
$^{***}$ $p < .001$.
$^{**}$ $p < .01$.
$^{*}$ $p < .05$.
$^{\dagger}$ $< 0.10$.

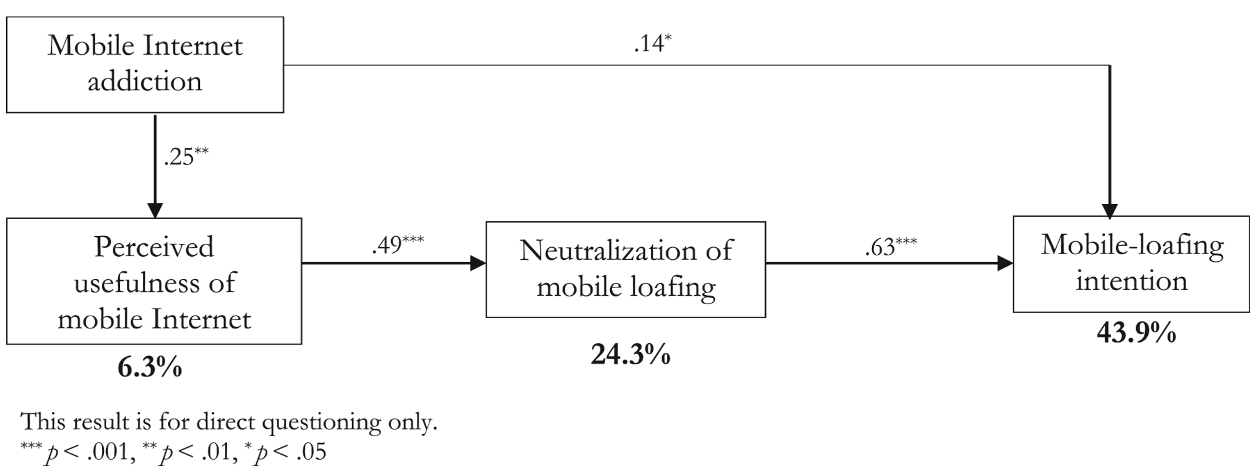


**Fig. 2.** Results of uncorrected structural model.

**Table 2**
Differences between indirect and direct questioning.

| Source | Multivariate test (Wilks' λ) | | | ANOVA test | | | | Result |
|---|---|---|---|---|---|---|---|---|
| | Value | F | Sig | Dependent variable | Mean Square | F | Sig | |
| Questioning type | .76 | 32.51 | .000 | Mobile Internet addiction | 230.63 | 113.02 | .000 | Biased |
| | | | | Perceived usefulness of mobile Internet | .12 | .05 | .821 | Unbiased |
| | | | | Neutralization of mobile loafing | 29.87 | 11.18 | .001 | Biased |
| | | | | Mobile-loafing intention | 19.03 | 6.18 | .013 | Biased |

results of correlations between the study's constructs and two BIDR subscales, including the SDE and the IM scales. In indirect questioning, perceived usefulness, neutralization, and mobile-loafing intention have insignificant correlations with the SD scales, suggesting these constructs are less susceptible to SD bias. However, indirect questioning of mobile Internet addiction has significant correlations with the IM scale,

**Table 3**
Correlations between study constructs and SD scales.

| Construct | | SDE | IM | Result |
|---|---|---|---|---|
| Indirect questioning | Mobile Internet addiction | −0.108 | −0.180* | Biased |
| | Perceived usefulness of mobile Internet | −0.051 | −0.033 | Unbiased |
| | Neutralization of mobile loafing | −0.037 | .073 | Unbiased |
| | Mobile-loafing intention | .059 | −0.119 | Unbiased |
| Direct questioning | Mobile Internet addiction | −0.333*** | −0.325*** | Biased |
| | Perceived usefulness of mobile Internet | .020 | −0.007 | Unbiased |
| | Neutralization of mobile loafing | −0.083 | −0.127† | Biased |
| | Mobile-loafing intention | −0.075 | −0.211** | Biased |

Notes:.
A two-tailed test was used.
*** $p < .001$,.
** $p < .01$,.
* $p < .05$,.
† $p < .10$.

implying some degree of SD bias.[4] The results of correlations between direct questioning and SD scales suggest that mobile Internet addiction, neutralization, and mobile-loafing intention are susceptible to SD bias, but SD bias is not a significant influence on perceived usefulness.

### 5.3. Factor mixture model

Finally, based on Leite and Cooper [89], we used Mplus 8 to conduct an FMM for each combination of a focal construct and an SD scale (i.e., SDE or IM). In total, we conducted eight sets of FMM (4 constructs × 2 SD scales). First, we report the results using IM as an example condition (see summary in Table 4; details in Table 5). In essence, FMM detected SD bias in mobile Internet addiction and mobile-loafing intention but not in neutralization and perceived usefulness. However, we discarded FMM as no longer useful in further profiling for SD bias. This decision came after the Lo–Mendell–Rubin (LMR) and adjusted LMR (aLMR) tests consistently indicated that, assuming measurement invariance of the factor loadings, a two-class or above-two-class mixture model did not fit the data statistically any better than a one-class model (LMR and aLMR columns: all $p > .05$). But it is noteworthy that the results of the various information criteria (i.e., Akaike information criterion (AIC), Bayesian information criterion (BIC), sample size-adjusted BIC (aBIC)) could provide alternative interpretations. We also performed a series of diagnostic checks to confirm that the measurement invariance of the factor loadings was a reasonable assumption in our context and data. The pattern of results contrasts with that of Leite and Cooper [89], implying that mixture models benefit neither the context of mobile loafing, the specific scales, nor subject population of this study.

**Table 4**
Summary results of factor mixture model.

| Construct | SDE | IM |
|---|---|---|
| Mobile Internet addiction | SD detected (One class with cross-loading) | SD detected (One class with cross-loading) |
| Perceived usefulness of mobile Internet | SD not detected (One class no cross-loading) | SD not detected (One class no cross-loading) |
| Neutralization of mobile loafing | SD not detected(One class no cross-loading) | SD not detected (One class no cross-loading) |
| Mobile-loafing intention | SD detected(Three classes heterogeneous with cross-loading) | SD detected (One class with cross-loading) |

Although FMM may have been of little use in our study, its results nevertheless revealed the possible presence of SD bias. Comparison of the factor analysis results in terms of information criteria (i.e., AIC, BIC, aBIC) revealed the better fit of a model in which the items of the focal scale had cross-loadings on the latent SD bias factor. This was true for both mobile Internet addiction and mobile-loafing intention and thus, in each case, indicated detection of SD bias. (Mobile Internet Addiction Model No. 2 in Table 5 was smaller than Model No. 1 across all criteria; moreover, Table 5's Mobile Loafing Intention Model No. 2 had smaller AIC and aBIC.) However, SD bias was not detected in either perceived usefulness or neutralization because a model without cross-loadings of the focal items on the SD bias factor had a better fit with data in perceived usefulness and neutralization. (Table 5's Perceived Usefulness Model No. 2 did not even converge, and Table 5's Neutralization Model No. 1 consistently had smaller information criteria.)

### 5.4. Summary of detection methods

As expected, the results of the three detection methods show that mobile Internet addiction and mobile-loafing intention were contaminated by SD bias. However, perceived usefulness was free of SD bias. Also, comparisons with indirect questioning and correlation with SD scales show a certain degree of SD bias in neutralization, but an FMM did not detect SD bias in it. As we noted earlier, an FMM may not be useful in assessing SD bias. In sum, our study examines mobile Internet addiction, neutralization, and mobile-loafing intention as biased constructs and perceived usefulness as an unbiased construct to examine the significance of SD bias in causal relationships.

## 6. Correction via partial correlation

### 6.1. Variance reduction rate

Using partial correlations, we examined the VRR of four relationships under three conditions: (1) biased independent variable and unbiased dependent variable, (2) unbiased independent variable and biased dependent variable, and (3) biased independent variable and biased dependent variable. Table 7 shows results of the VRR. Although the VRR is not a tool for assessing the extent of SD bias, it is useful to determine how many pairs of variables have a correlation that can be attributed to an unmeasured variable (i.e., SD in our study). For example, Chen and Spector [91] examined the extent to which negative affectivity inflated correlations between chronic job stressors and strains. Using the VRR, they found that negative affectivity explained a large amount of shared variance (56%−100%) between stressor and physical strains such as absence, physical symptoms, and doctor visits.

The biased independent variable and unbiased dependent variable include the relationship (a) between mobile Internet addiction and perceived usefulness ($r_{xy} = 0.243$). The maximum VRR was −19.8% from second-order partial correlation ($r_{xy.SD} = 0.266$). The unbiased independent variable and biased dependent variable condition examines the relationship (b) between perceived usefulness and neutralization ($r_{xy} = 0.482$). We found partial correlation ($r_{xy.SD} = 0.486$) has almost no impact with −1.7% of the VRR. Finally, the biased independent and dependent variable conditions permitted examination of the relationships (c) between neutralization and mobile-loafing intention ($r_{xy}$ =

[4] There are two plausible explanations why indirect questioning of mobile Internet addiction contains SD bias: (1) A respondent may project himself or herself onto typical Internet users, and (2) the degree of SD bias in mobile Internet addiction is so strong that even the general indirect question can invoke a desire to answer in a socially desirable manner.

**Table 5**
Detailed results of factor mixture model.

| SD | Construct | Model # | AIC | BIC | aBIC | LMR*p* value | aLMR*p* value |
|---|---|---|---|---|---|---|---|
| SDE | Mobile Internet addiction | 1. One class no cross-loading | 4634.3 | 4753.9 | 4639.9 | | |
| | | 2. One class with cross-loading | **4620.3** | **4753.2** | **4626.5** | | |
| | | 3. Two-class mixture model | 4568.1 | 4714.3 | 4574.9 | 0.23 | 0.23 |
| | Perceived usefulness of mobile Internet | 1. One class no cross-loading | **3797.6** | **3907.3** | **3802.7** | | |
| | | 2. One class with cross-loading | —† | —† | —† | | |
| | | 3. Two-class mixture model | 3794.9 | 3927.9 | 3801.1 | 0.23 | 0.23 |
| | Neutralization of mobile loafing | 1. One class no cross-loading | **3825.2** | **3934.8** | **3830.3** | | |
| | | 2. One class with cross-loading | 3829.3 | 3948.9 | 3834.9 | | |
| | | 3. Two-class mixture model | 3789.8 | 3922.7 | 3796.0 | 0.23 | 0.23 |
| | Mobile-loafing intention | 1. One class no cross-loading | 3585.9 | 3695.6 | 3591.0 | | |
| | | 2. One class with cross-loading | 3589.8 | 3709.5 | 3595.4 | | |
| | | 3. Two-class mixture model | 3527.1 | 3660.0 | 3533.2 | **0.00** | **0.00** |
| | | 4. Three-class mixture model | **3496.0** | **3638.9** | **3502.6** | **0.00** | **0.00** |
| IM | Mobile Internet addiction | 1. One class no cross-loading | 4781.4 | 4901.0 | 4786.9 | | |
| | | 2. One class with cross-loading | **4766.4** | **4899.3** | **4772.6** | | |
| | | 3. Two-class mixture model | 4715.3 | 4861.5 | 4722.1 | 0.23 | 0.23 |
| | Perceived usefulness of mobile Internet | 1. One class no cross-loading | **3944.7** | **4054.3** | **3949.8** | | |
| | | 2. One class with cross-loading | —† | —† | —† | | |
| | | 3. Two-class mixture model | 3929.7 | 4062.6 | 3935.9 | 0.23 | 0.23 |
| | Neutralization of mobile loafing | 1. One class no cross-loading | **3972.2** | **4081.9** | **3977.3** | | |
| | | 2. One class with cross-loading | 3972.5 | 4092.1 | 3978.1 | | |
| | | 3. Two-class mixture model | 3939.8 | 4072.7 | 3945.9 | 0.23 | 0.23 |
| | Mobile-loafing intention | 1. One class no cross-loading | 3733.0 | **3842.7** | 3738.1 | | |
| | | 2. One class with cross-loading | **3727.0** | 3846.6 | **3732.6** | | |
| | | 3. Two-class mixture model | 3665.9 | 3798.9 | 3672.1 | 0.23 | 0.23 |

Notes: A one-class FMM reduces to a confirmatory factor analysis, thus LMR and aLMR tests are unavailable. † Model analysis did not converge with various starting values. Model indices were thus unavailable. AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; aBIC = sample size-adjusted BIC; LMR *p* value = *p*-value of the LMR test [90]; aLMR *p* value = *p*-value of the aLMR test [90].

**Table 6**
Summary of detection methods.

| Construct | Comparison with indirect questioning | Correlation with SD scales | Factor mixture model | Conclusion of our study |
|---|---|---|---|---|
| Mobile Internet addiction | Biased | Biased | Biased | Biased |
| Perceived usefulness of mobile Internet | Unbiased | Unbiased | Unbiased | Unbiased |
| Neutralization of mobile loafing | Biased | Biased | Unbiased | Biased |
| Mobile-loafing intention | Biased | Biased | Biased | Biased |

0.643) and (f) between mobile Internet addiction and mobile-loafing intention ($r_{xy} = 0.219$). Partial correlations result in adjusting a maximum of 2.2% of the VRR in (c) ($r_{xy.SD} = 0.636$) and 46.0% of the VRR in (d) ($r_{xy.SD} = 0.161$) from IM-controlled partial correlation. The notable amount of the variance reduction in (d) may be the result of moderate SD bias contaminating the independent ($r_{xSD} = -0.325$) and dependent ($r_{ySD} = -0.211$) variables. It is important to note that although both the independent and dependent variables are susceptible to SD bias, the SD did not account for much of the variance shared by neutralization and mobile-loafing intention ($r_{xSD} = -0.211$). One plausible explanation is that neutralization is contaminated by the low level of SD bias ($r_{ySD} = -0.127$, $p = .082$).

### 6.2. SD corrected structural models

We conducted structural equation modeling by using first- and

**Table 7**
Results of the variance reduction rate.

| SD bias conditionIV-DV | Independent variable | Dependent variable | Correlation | *r* | Variance | Variance reduction rate |
|---|---|---|---|---|---|---|
| (a)Biased-Unbiased | Mobile Internet addiction | Perceived usefulness of mobile Internet | Zero orderFirst orderSDE controlledIM controlledSecond order[a] | .243**.264**.254**.266** | 5.9%7.0% 6.5%7.1% | −18.0%−9.3%−19.8% |
| (b) Unbiased-Biased | Perceived usefulness of mobile Internet | Neutralization of mobile loafing | Zero orderFirst orderSDE controlledIM controlledSecond order | .482***.486***.485***.486*** | 23.2%23.6% 23.5%23.6% | −1.7%−1.3%−1.7% |
| (c)Biased-Biased | Neutralization of mobile Loafing | Mobile-loafing intention | Zero orderFirst orderSDE controlledIM controlledSecond order | .643***.641***.636***.637*** | 41.3%41.1% 40.4%40.6% | 0.6%2.2% 1.9% |
| (d)Biased-Biased | Mobile Internet Addiction | Mobile-loafing intention | Zero orderFirst orderSDE controlledIM controlledSecond order | .219**.206**.161*.171* | 4.8%4.2% 2.6%2.9% | 11.5%46.0% 39.0% |

Notes:.
A two-tailed test was used.
*** $p < .001$,.
** $p < .01$,.
* $p < .05$.
[a] both SDE and IM controlled.

second-order partial correlations. As shown in Table 8, goodness of fit indices of all SD corrected models showed good model fits. Consistent with the results of the VRR, a notable correction was made in the IM-corrected model and in both the SDE- and IM-corrected model. For example, in the IM-corrected model, a path coefficient and its significance were changed from 0.14 ($p < .05$) to 0.10 ($p > .05$) (see Fig. 3). As mentioned earlier, this change can result from the influence of SD bias on both the independent and dependent variables. Accordingly, SD bias may lead to a false-positive result because of the change in statistical significance when both the independent and dependent variables are contaminated by SD bias.

## 7. Discussion and conclusions

### 7.1. Summary of the study

The objective of this study was to assess the extent of SD bias and find an effective way to control, if any, such bias in IS research. We conducted a survey study to explore how seriously SD bias could distort our inferences and to examine the efficacy of partial correlations in reliably estimating causal relationships. In particular, we studied mobile loafing based on data collected from 205 survey participants. The survey results showed that even when SD bias contaminates either an independent or a dependent variable, it is not a severe problem [13,15]. When we used indirect questioning to test for SD bias, the results remained the same as with direct questioning. However, when both the independent and dependent variables are prone to SD bias, its presence can become a real threat to the validity of research on sensitive topics [6]. The findings and additional suggestions of our study are summarized in Table 9. Overall, this study contributes to IS research by demonstrating the potential damage of SD bias and the usefulness of alternative remedial measures against it through multiple detection methods.

### 7.2. Theoretical and methodological contributions

First, prior research examined SD bias in only an independent variable condition and concluded that the condition was not problematic [25]. Extending Paunonen and LeBel [25], our empirical survey contributes to the literature on SD bias by adding two more conditions: SD bias in only a dependent variable and in both independent and dependent variables simultaneously. Our overall results are similar to those of Paunonen and LeBel [25] when only one variable (independent or dependent variable) is susceptible to SD bias. However, when both variables are contaminated, SD bias led to inflating biases in estimation. This finding is unique and contributes to the literature because, although the direction of bias in estimated relationships is downward when only one variable contains SD bias [25], it becomes upward distortion when both variables are biased. Furthermore, our theoretical contribution lies in showing that, according to our findings, researchers should no longer ignore the potential threat SD bias poses to empirical-based analyses and conclusions. Researchers who fail to consider the possibility of SD bias in both the independent and dependent variables may be inclined to accept their research hypotheses erroneously, hence overclaiming the validity of their theoretical model and contribution.

**Table 8**
Goodness of fit indices.

| Model | $\chi^2$ (DF) | $\chi^2$/ DF | CFI | GFI | AGFI | SRMR | RMSEA |
|---|---|---|---|---|---|---|---|
| Uncorrected | 90.225 (61) | 1.479 | .988 | .933 | .900 | .061 | .048 |
| SDE corrected | 91.349 (61) | 1.498 | .987 | .932 | .898 | .062 | .049 |
| IM corrected | 91.046 (61) | 1.493 | .987 | .933 | .900 | .065 | .049 |
| SDE and IM corrected | 92.514 (61) | 1.517 | .987 | .932 | .898 | .066 | .050 |

Second, our findings from the survey can resolve questions surrounding the inconsistent effects of the extent of SD bias in prior IS research. Although Turel et al. [14] and Soror et al. [13] argued that SD bias was not a problem in their studies, Kwan et al. [6] found that SD bias significantly inflates the coefficients of structural paths, suggesting that SD bias is problematic. These inconsistent conclusions are because of different SD bias conditions. For example, Turel et al. [15] had only one sensitive variable (e.g., online auction addiction), but Kwan et al. [6] had multiple sensitive variables (e.g., punishment certainty, software piracy attitude, and behavior). By assessing the extent of SD bias under different conditions, we found that only a two-biased-variable condition is a serious concern. Therefore, our survey results may reconcile the different conclusions of the effects of SD bias in prior IS research.

Third, although prior IS research generally relied on a single method to detect SD bias (e.g., SD scale), our study is among the first in IS research to use multiple methods (i.e., a comparison with indirect questioning, correlation with an SD scale, and an FMM) to further assess SD bias. Our study showed that an FMM [89] is less sensitive in detecting SD bias than the other two methods. Because comparisons with indirect questioning can detect only the existence of SD bias, SD scales are more useful than this technique because they can assess the level of SD bias. Furthermore, SD scales in conjunction with partial correlations can be used to control for SD bias when SD bias is found. Unlike other alternative techniques for preventing SD bias, for example, the RRT [6,20] and the crosswise model [48], SD scales are readily compatible with the conventional method of survey (or online survey) data collection. A researcher has only to create one or two additional blocks of survey questions without having to restructure the entire survey or introduce unnecessary scenarios that might confuse or distract a respondent's response process.

Finally, this study found the IM scale useful in measuring and controlling SD bias in the context of mobile loafing. Our study found that the IM scale outperformed the SDE scale. Recent research emphasized that it is essential to distinguish between egoistic bias and moralistic bias in a self-favoring response tendency. These two can be well-measured by the SDE and IM subscales of the BIDR, respectively [7,35,41]. Egoistic bias occurs when people emphasize individuality, personal striving, uniqueness, and accomplishment; moralistic bias arises when people highlight relationships, intimacy, affiliation, and benefiting others and society [35,41]. The IM scale is appropriate in negative IT use (IT addiction, cyberbullying, security policy violations, digital piracy, and cyberloafing) because those contexts are socially deviant and can bring moralistic bias. Furthermore, Uziel [93] noted that the IM scale should be a measure of "interpersonally oriented self-control that identifies individuals who demonstrate high levels of self-control, especially social context" (p. 243). In sum, the IM scale is more efficient to administer than the complete BIDR scale and appropriate in negative IT use in recovering unbiased estimates in the context of negative IT use.

### 7.3. Practical contributions

This research provides practitioners with useful insights into how to handle SD bias in their survey studies. This study shows that SD bias does not contaminate the estimate of a causal relationship when it affects only one of the independent and dependent variables. Thus, if only one of the variables in question is sensitive, there may be little need to include the rather costly, time-consuming, 16-item BIDR scale in a survey questionnaire. Even without the BIDR, the relationship between the independent and dependent variables is shown to be reliably recovered in this study. In contrast, when SD bias affects both the independent and dependent variables simultaneously, an SD scale like the BIDR should be included to control SD bias in estimating their relationship. However, we found that in recovering unbiased estimates, the 8-item IM scale was the equal of the 16-item BIDR that also includes the

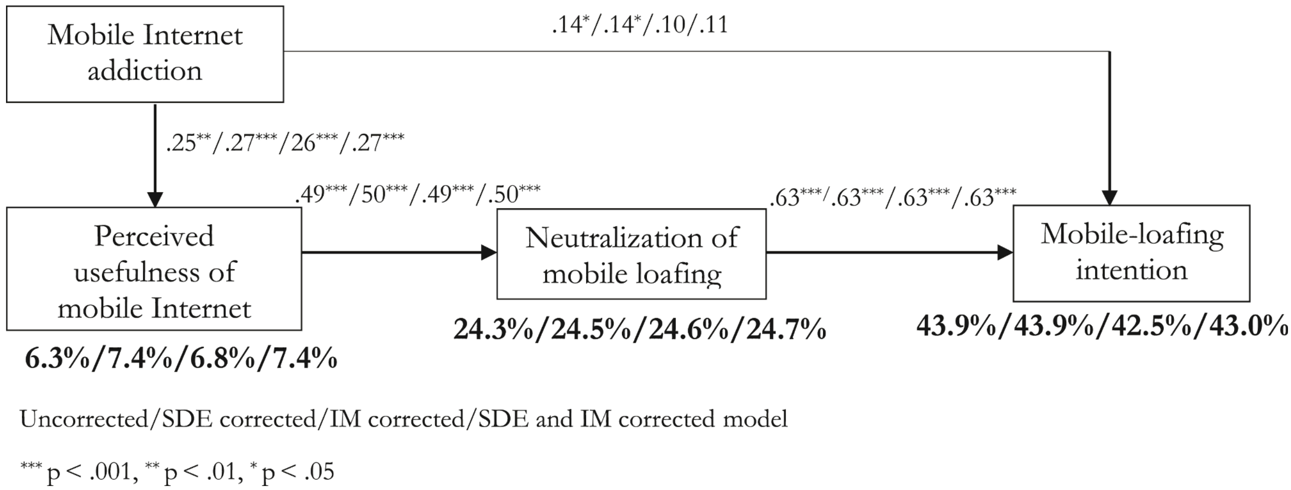


**Fig. 3.** Results of SD-corrected structural models.

**Table 9**
Summary of findings.

| Findings | Additional Suggestions |
|---|---|
| Two detection methods (i.e., comparison with indirect questioning and correlation with SD scales) show that mobile Internet addiction, neutralization, and mobile-loafing intention are susceptible to SD bias, but perceived usefulness is free of it.<br>However, a factor mixture model does not detect SD bias in neutralization, and it is relatively ineffective in assessing SD bias. | Indirect questioning is not directed toward a specific subject under study but instead toward people in general. Although indirect questioning can be used to identify the existence of SD bias, we do not suggest use of the technique for preventing SD bias. |
| SD bias does not seriously contaminate estimates of causal relationships when SD bias influences only one variable (either independent or dependent variable).<br>SD bias is less problematic when both the independent and dependent variables are contaminated by a low level of SD bias.<br>However, SD bias may threaten estimates of causal relationships when both the independent and dependent variables are contaminated by at least a moderate level of SD bias. | We focused on the extent of SD bias in situations in which the independent and/or dependent variable is susceptible to SD bias in the context of mobile loafing. Because the degree of SD bias generally depends on the context and contingencies of admitting to undesirable behavior, future research needs to examine our findings in more serious contexts such as cyber sexual harassment. |
| The VRR shows that relative correction is larger when a correlation is smaller ($r_{xy} = 0.219$) than larger ($r_{xy} = 0.643$).<br>Partial correlation corrects a maximum of 46.0% of the variance of SD bias-prone correlation ($r_{xy \cdot SD} = 0.161$ vs. $r_{xy} = 0.219$).<br>In an IM-corrected structural model, the path coefficient and its significance were changed from 0.14 ($p < .05$) to 0.10 ($p > .05$).<br>The IM scale is an effective and pragmatic choice in assessing SD bias and in recovering unbiased estimates of causal relationships in the context of problematic behaviors. | Because the IM scale is appropriate in moralistic contexts [7], we used and recommended using it in contexts such as cyberloafing, cyberbullying, and technology addiction. However, the IM scale still has some limitations. For example, it confounds trait honesty with response bias [92]. Thus, our findings need to be reexamined using different SD scales or methods. |

8-item SDE scale. Thus, practitioners may consider using the IM over the BIDR when they need to reduce the length of a survey questionnaire. Our finding suggests that the inferences they will acquire are likely to be comparable regardless of whether they choose the IM scale or the BIDR.

### *7.4. Limitations and further research directions*

Some limitations of this study should be mentioned. First, our survey-based study used data specific to mobile loafing (e.g., [94]) and IT addiction (e.g., [15,17,61]). Because mobile loafing and IT addiction are not as serious as other contexts, such as Internet drug trafficking and cyber sexual harassment, the results could differ if more serious contexts were examined. Second, as with other research on SD bias, this study assumes that individuals' responses to an SD scale are reliable. However, people may not answer SD-related items truthfully; our estimate of SD bias could be distorted in such a case. Third, another assumption of this study was that SD bias emerges from a single source of SD. Multiple sources of SD could simultaneously bias a specific research variable, a complicated scenario not considered by this study or by other research on SD bias. Fourth, on a similar note as the previous point, research variables can be contaminated by SD bias and other types of biases such as common method bias. Thus, our findings should be interpreted carefully until we have a fuller understanding of how other potential sources of biases may interact with SD bias in regulating individuals' responses to measurement items. Fifth, this study concluded that the IM scale is proper in the context of negative IT use that brings about moralistic bias [3,7]. However, Müller and Moshagen [92] demonstrated that the IM scale confounds trait honesty with response bias. Thus, our findings need to be retested using different SD scales or methods. Another potential limitation relates to the nature of mobile addiction. This study shows that a structural relationship is distorted when the two constructs (e.g., the addiction-intention path) are highly contaminated by SD bias. However, the distorted relationship found in our study was associated only with mobile Internet addiction. Thus, we cannot exclude the possibility that our findings are unique to the unknown property of mobile Internet addiction but not necessarily because of the reason we proposed.

This study opens exciting and fertile avenues for further research. First, a natural extension of this study would be to consider multiple antecedents and outcome variables. Specifically, it remains to be seen whether our findings apply to a situation in which multiple independent variables are correlated. Similarly, further research should examine the case of correlated outcomes. Second, although there are several rational techniques to reduce SD bias, we introduced indirect questioning, the RRT, and implicit measures because these techniques were used in IS research. Future IS research can examine other techniques such as bogus pipeline techniques [95] and item-specific social-desirability rating [96]. Finally, future research needs to explore the different effects of the SDE scale and the IM scale on research variables to understand SD bias. The SDE scale and the IM scale represent different aspects of SD bias [3, 4,7,39], and they are supposed to contaminate research variables in different ways. Our inferences are likely to vary depending on how the SDE scale and the IM scale differentially affect independent and dependent variables. Thus, researchers are encouraged to study how these two distinct sources of SD bias in a two-biased-variable scenario distort our inferences.

### 7.5. Concluding remarks

Social desirability bias has become increasingly crucial in IS research because IT use has expanded to encompass a broader range of sensitive domains such as software piracy, online gambling, cyberloafing, and cyberbullying. Despite this increasing concern, little was known about how serious SD bias might be in estimating causal relationships, and more importantly, how to effectively and efficiently correct for such bias. Through a survey in the context of mobile loafing, this study demonstrates that SD bias could be problematic when both independent and dependent variables are contaminated. Yet, it also shows that the application of an SD scale and partial correlation are beneficial in recovering true relationships free of SD bias. We hope that more research will be directed toward this crucial topic of SD bias and that the systematic approach used in this study will serve as a basis for such endeavors.

## CRediT authorship contribution statement

**Dong-Heon (Austin) Kwak:** Conceptualization, Data curation, Formal analysis, Writing – review & editing. **Xiao Ma:** Formal analysis, Funding acquisition. **Sumin Kim:** Writing – review & editing.

## Declaration of Competing Interest

None.

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## Appendix: Instruction and measurement

Mobile loafing, or “nonwork-related mobile computing,” refers to employees’ using mobile Internet during business hours for personal reasons (e.g., gaming, browsing websites, sending messages, etc.). Please answer the following questions.

1. *Direct questioning*

*1.1. Mobile Internet addiction [15,82]*

ADDd1: My social life has sometimes suffered because of me interacting with mobile Internet.

ADDd2: When I am not using mobile Internet, I often feel agitated.

ADDd3: I have made unsuccessful attempts to reduce the time I interact with mobile Internet.

ADDd4: I often fail to get enough rest because of my use of mobile Internet.

*1.2. Perceived usefulness of mobile Internet [66]*

PUd1: I find mobile Internet useful at work.

PUd2: Using mobile Internet helps me accomplish things more quickly at work.

PUd3: Using mobile Internet increases my productivity at work.

*1.3. Neutralization of mobile loafing [29]*

I think that

NEUd1: it is acceptable to engage in mobile loafing if no harm is done to the company.

NEUd2: hard work can compensate for engaging in mobile loafing.

NEUd3: if a person gets the job done, it is all right to occasionally engage in mobile loafing.

*1.4. Mobile-loafing intention [29]*

INTd1. I predict that I would use mobile Internet at work for non-work–related purposes in the future.

INTd2. I intend to use mobile Internet at work for nonwork-related purposes in the future.

INTd3. I plan to use mobile Internet at work for nonwork-related purposes in the future.

2. *Indirect questioning*

*2.1. Mobile Internet addiction [82]*

ADDi1: A typical mobile Internet user’s social life has sometimes suffered because of him/her interacting with mobile Internet.

ADDi2: When a typical mobile Internet user is not using mobile Internet, he/she often feels agitated.

ADDi3: A typical mobile Internet user has made unsuccessful attempts to reduce the time he/she interact with mobile Internet.

ADDi4: A typical mobile Internet user often fails to get enough rest because of his/her use of mobile Internet.

*2.2. Perceived usefulness of mobile Internet [66]*

PUi1: A typical mobile Internet user finds mobile Internet useful at work.

PUi2: Using mobile Internet helps a typical mobile Internet user accomplish things more quickly at work.

PUi3: Using mobile Internet increases a typical mobile Internet user’s productivity at work.

*2.3. Neutralization of mobile loafing [29]*

A typical mobile Internet user thinks that

NEUi1: it is acceptable to engage in mobile loafing if no harm is done to the company.

NEUi2: hard work can compensate for engaging in mobile loafing.

NEUi3: if a person gets the job done, it is all right to occasionally engage in mobile loafing.

*2.4. Mobile-loafing intention [29]*

INTi1. A typical mobile Internet user predicts that he/she would use mobile Internet at work for nonwork-related purposes in the future.

INTi2. A typical mobile Internet user intends to use mobile Internet at work for nonwork-related purposes in the future.

INTi3. A typical mobile Internet use plans to use mobile Internet at work for nonwork-related purposes in the future.

3. *Balanced inventory of desirable responding-16*

Please answer the following questions with respect to yourself (1: Not true – 7: Very true)

*3.1. Self-deception enhancement [3]*

SDE1: I have not always been honest with myself. (R)

SDE2: I always know why I like things.

SDE3: It’s hard for me to shut off a disturbing thought. (R)

SDE4: I never regret my decisions.

SDE5: I sometimes lose out on things because I can not make up my mind soon enough. (R)

SDE6: I am a completely rational person.

SDE7: I am very confident of my judgments.

SDE8: I have sometimes doubted my ability as a lover. (R)

Add one point for every “6″ or “7″ (minimum = 0: maximum = 8)

(R) Items keyed in the “false” (negative) direction.

*3.2. Impression management [3]*

IM1: I sometimes tell lies if I have to. (R)

IM2: I never cover up my mistakes.

IM3: There have been occasions when I have taken advantage of someone. (R)

IM4: I sometimes try to get even rather than forgive and forget. (R)

IM5: I have said something bad about a friend behind his or her back. (R)

IM6: When I hear people talking privately, I avoid listening.

IM7: I never take things that do not belong to me.

IM8: I do not gossip about other people's business.

Add one point for every "6″ or "7″ (minimum = 0: maximum = 8)

(R) Items keyed in the "false" (negative) direction.

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